BUSINESS REPORT: COMPARATIVE ANALYSIS OF SHELL PLC AND JSC NC KAZMUNAYGAS

Course: Business Administration (BA2026) · Year 1, Trimester 1
Assignment: Assignment 1 - Business Report
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Group: DPA-2602
Company A: Shell
Company B: KazMunayGas (KMG)
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Business Context and Economic Foundations

JSC NC "KazMunayGas" (KMG) is the state-owned vertically integrated oil and gas champion of the Republic of Kazakhstan, operating across the entire energy value chain including exploration, production, transportation, refining, and petrochemicals [3], [5]. In 2025, KMG achieved a consolidated oil and gas condensate production volume of 26.211 million tones representing 26% of Kazakhstan's total national hydrocarbon output [2]. In crude transportation, KMG conveyed 83 million tones (55% of the national total) across 8.9 thousand kilometers of trunk pipelines and marine routes [3], [10]. Refining intake at domestic and European facilities reached 21.0 million tones, supplying 80% of Kazakhstan's domestic fuel market [3], [7]. KMG's 2025 financial results demonstrated strong operational resilience: revenue grew by 12.5% to 9,371 billion KZT ($17,977 million), EBITDA reached 2,393 billion KZT ($4,591 million), and net profit totaled 1,072 billion KZT ($1,996 million) [3]. Net debt declined to a historical low of 375 billion KZT), maintaining an exceptionally conservative Net Debt-to-EBITDA ratio of 0.16x. The Group employs over 52,000 workers across its consolidated operating footprint [3].

In contrast, comparative benchmark Shell plc represents a global transnational energy major operating in open international capital and commodity markets [2], [4]. In 2025, Shell's total hydrocarbon production stood at 2.800 million barrels of oil equivalent per day [4]. Shell generated $18.5 billion in Adjusted Earnings, $42.9 billion in Cash Flow from Operations (CFFO), and $26.1 billion in Free Cash Flow (FCF) on cash capital expenditures of $20.9 billion [4]. Shell's capital allocation strategy prioritizes shareholder returns, delivering $22.4 billion (52% of CFFO) in 2025 through $8.5 billion in dividends and $13.9 billion in share buybacks, alongside cumulative structural cost reductions of $5.1 billion since 2022 [4]. While Shell shapes its portfolio around global LNG growth (targeting 4-5% annual sales growth) and deepwater high-margin barrels, KMG operates within Kazakhstan's socially oriented market economy, where commercial goals are legally bound to national energy security [1], [3], [5].

Environment, Stakeholders, Sustainability and Ethics

KMG's macro-environment is governed by its distinct ownership structure and quasi-sovereign legal status [1], [3]. The sovereign wealth fund Samruk-Kazyna JSC holds 67.42% of KMG's shares, the Ministry of Finance RK holds 20.00%, the National Bank RK holds 9.58%, and a 3.00% free float trades on KASE and AIX exchanges [3], [8]. The primary regulatory and ethical tension facing KMG is its dual mandate: generating commercial returns for public shareholders while fulfilling socio-economic obligations to the Kazakh state [1], [3]. KMG is mandated to supply crude oil to domestic refineries (ANPZ, PNHZ, PKOP) at subsidized domestic prices significantly below export netback parity, ensuring fuel price stability and domestic energy security [3], [5], [9].

In environmental sustainability, KMG executes its Low-Carbon Development Program to 2060, targeting a 15% absolute CO2 reduction by 2031 (vs 2019 baseline) [3]. KMG achieves a 98.8% associated petroleum gas (APG) utilization rate (flaring intensity of 1.70 t per 1,000 t

hydrocarbon production), reports methane emissions to UNEP under the OGMP 2.0 initiative (Gold Standard tracking), and is constructing a 247 MW hybrid solar/wind/gas power plant in Zhanaozen with Eni (50 MW solar commissioned in 2025) [3]. In the Caspian Sea, KMG performs acoustic vessel monitoring to protect marine fauna and maintains specialized oil spill response capabilities via KMG Systems & Services [3]. In contrast, Shell targets Net-Zero operational emissions by 2050 (achieving a 36% Scope 1&2 reduction by 2025 vs 2016) and eliminated routine upstream flaring in 2025 [4]. However, while Shell aggressively divests carbon-intensive assets (e.g., selling SPDC in Nigeria), KMG must maintain mature, labor-intensive onshore assets in Mangystau and Atyrau to preserve regional employment [3], [5].

Business Model, Ownership and Digital Transformation (AI)

KMG's business model relies on state ownership that secures statutory priority subsoil use rights via direct negotiations and direct equity participation in mega-consortia (Tengiz 20%, Kashagan 16.88%, Karachaganak 10%) [3], [5]. However, state ownership restricts capital flexibility by obligating KMG to finance regional social infrastructure, such as the 50,000 m^3/day Kenderli seawater desalination plant (121.2 billion KZT) and the new Zhanaozen Gas Processing Plant [3].

Digital transformation and Artificial Intelligence serve as essential drivers of operational efficiency under Kazakhstan's national "Digital Qazaqstan" strategy [9], [11]. KMG has deployed its proprietary ABAI Information System (an enterprise Big Data and AI platform developed by KMG Engineering) [3]. ABAI's "Waterflooding Management" AI module spans 1,660 injection cells across OMG, MMG, and KOA assets [3]. For occupational safety, KMG implemented the TUMAR computer vision AI system across 60 well workover crews, automatically detecting safety violations in real time [3]. At the Uzen field, KMG scaled Slim Hole drilling technology, reducing well construction CAPEX by 30% (4.6 billion KZT savings) [4]. At refineries, KMG is deploying predictive failure analytics to transition to 3-year extended turnaround cycles (UMRP) [3], [9]. Comparatively, Shell invests $1.17 billion in R&D, utilizing the T-Pulse AI safety vision system (reducing safety risks by 70%), AI-automated deepwater drilling at the Whale platform, and immersion cooling fluids for AI data centers [4].

Comparative Analysis and Transfer Judgments

Directly adopting Shell's strategy would compromise KMG's national mandate [1]. Transfer of technologies and managerial practices requires a structured, nuanced framework with explicit transfer judgments:

1. Adopt: AI Predictive Maintenance & Computer Vision Safety Systems
Reasoning: KMG should fully adopt Shell's practice of enterprise-wide predictive failure analytics and CCTV computer vision safety monitoring (modeled on Shell's T-Pulse). Extending TUMAR and ABAI across all domestic refineries (ANPZ, PNHZ, PKOP) will eliminate unplanned downtime and drastically reduce industrial accidents [3], [4].

2. Adapt: Climate Strategy & Decarbonization Framework
Reasoning: Shell's Net-Zero framework must be adapted to Kazakhstan's fossil-reliant economy. KMG should focus on Scope 1&2 methane abatement (OGMP 2.0 Gold Standard), hybrid renewable/gas generation (Zhanaozen 247 MW), and CCUS screening at gas processing units rather than divesting producing hydrocarbon assets [3], [4], [5].

3. Pilot: Digital Twin Reservoirs & Advanced EOR / Slim Hole Drilling
Reasoning: Shell's digital twin subsoil modeling and advanced EOR techniques should be piloted across declining mature fields (Embamunaygas, Ozenmunaygas) to optimize sweep efficiency and extend field life [3], [4].

4. Delay: Mass Capital Allocation to Green Hydrogen & EV Retail Infrastructure
Reasoning: Unlike Shell (which expands Shell Recharge and green hydrogen electrolysers), KMG should delay heavy CAPEX in green hydrogen and EV charging networks until EV fleet adoption and market demand mature in Central Asia, restricting current involvement to R&D desk studies [3], [4], [5].

5. Reject: Pure Market Pricing & Aggressive Divestment of Social Downstream Assets
Reasoning: Shell's strategy of shutting down or selling lower-margin refining assets must be explicitly rejected. Abandoning domestic fuel subsidies or selling domestic refineries would cause severe fuel shortages, economic inflation, and social disruption in Kazakhstan [1], [3], [5], [9].

Actionable Recommendations for KazMunayGas in Kazakhstan

1. Scale AI Predictive Analytics for Equipment Failure across ANPZ and PNHZ Refineries:
• Lead Owner: KMG Digital Transformation Department & Refinery Technical Directors.
• Action: Install continuous vibration/thermal sensor arrays and deploy machine learning failure prediction algorithms across critical refining units.
• Target KPI: Reduce unplanned refinery downtime by 15% and lower maintenance OPEX by 10% by end of 2027.

2. Expand Slim Hole Drilling & ABAI Waterflooding AI at Uzen Field:
• Lead Owner: KMG Engineering LLP & JSC Ozenmunaygas.
• Action: Drill 132 Slim Hole wells integrated directly with ABAI's AI Waterflooding Management module.
• Target KPI: Achieve 30% CAPEX savings per well (4.6B KZT savings) and unlock 380,000 tonnes/year of incremental crude production.

3. Deploy Integrated AI & Satellite Methane Risk Monitoring Platform for Caspian & Onshore Assets:
• Lead Owner: KMG Systems & Services LLP & HSE Department.
• Action: Fuse optical OGI camera data, aerial laser surveys, and satellite imagery into an automated "Methane Leak Atlas" under OGMP 2.0.

• Target KPI: Cut leak detection response times to 30 minutes and maintain APG flaring intensity below 1.5 t per 1,000 t output by 2027.

Reference List

[1] OpenStax, Introduction to Business 2e, Rice University, Houston, TX, USA, 2024. [Online]. Available: https://openstax.org

[2] KAZENERGY Association, National Energy Report 2023, Astana, Kazakhstan, 2023.

[3] JSC NC "KazMunayGas", Annual Report 2025: Global Standards. National Priorities, Astana, Kazakhstan, Apr. 28, 2026. [Online]. Available: https://www.kmg.kz

[4] Shell plc, Annual Report and Accounts 2025: Deliver More Value with Less Emissions, London, UK, Mar. 11, 2026. [Online]. Available: https://www.shell.com

[5] JSC NC "KazMunayGas", Consolidated Financial Statements for the Year Ended December 31, 2025, PwC Audit Report, Astana, Kazakhstan, Mar. 5, 2026.

[6] E. Akkenzhenov, Ministry of Energy Report on 2025 Performance and 2026 Digitalization Objectives, Ministry of Energy RK, Astana, 2026.

[7] KMG International N.V., KMGI Financial Performance and European Refining Operational Brief, Press Release, Bucharest, Romania, July 29, 2026.

[8] JSC NC "KazMunayGas", Corporate Governance and Shareholder Structure, Official Portal, 2026. [Online]. Available: https://www.kmg.kz

[9] Ministry of Energy of the Republic of Kazakhstan, Digitalization of Energy Sector and Refining Outlook, Official Report, Astana, 2026.

[10] Caspian Pipeline Consortium (CPC), Operational Crude Oil Transportation Volumes Brief, 2026.

[11] Administration of the President of the Republic of Kazakhstan, Address of the Head of State Kassym-Jomart Tokayev on AI and Digital Transformation, Official Site of the President of RK, 2026.

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ASSIGNMENT 1 APPENDIX: ARTIFACTS 1-3 & AI-USE DECLARATION

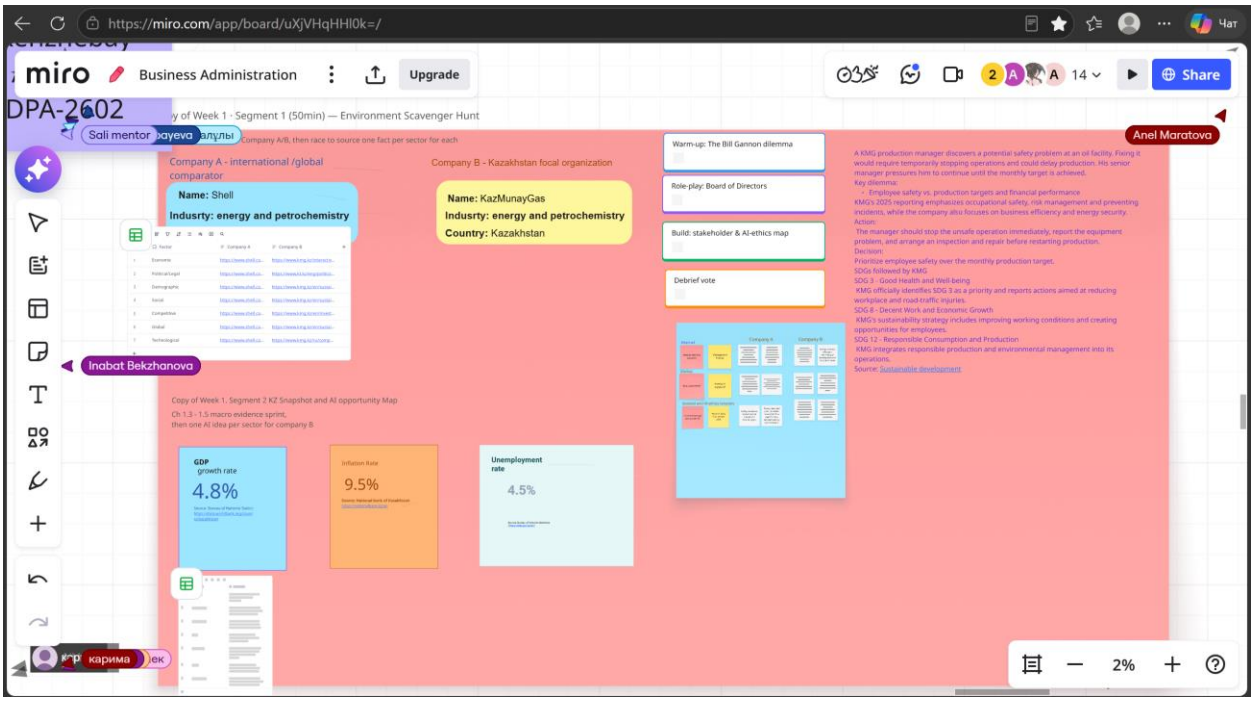
Course & Attestation: Business Administration (BA2026) -1st Attestation

Focal Company (Company B): KazMunayGas (KMG) - Astana, Kazakhstan

Comparative Company (Company A): Shell -London, United Kingdom

Student & Group: Kenzhebay Ayaulym DPA-2602

1. Artifact Week 1: Environment and AI-Opportunity Map for KazMunayGas



This matrix evaluates the external macro-environmental factors (PESTEL) affecting KazMunayGas in Kazakhstan, mapping local operational realities to digital and AI transformation opportunities grounded in course reading (OpenStax Ch. 1–3) and company data.

Macro-Environment Sector	Kazakhstan Operational & Market Reality (KMG)	Direct Strategic Impact on KMG	Potential Digital & AI Opportunity
Political / Legal & Regulatory	State mandate requiring crude supplies to domestic refineries (ANPZ, PNHZ, PKOP) at subsidized prices. Strict Tax Code regulations,	Subsidized crude limits domestic profit margins compared to export netback parity, creating a dual mandate tension.	AI-driven automated tax compliance tracking and economic yield

	oil export duties, and subsoil use contracts.		optimization between export and domestic refining allocations.
Economic & Market Dynamics	High revenue volatility linked to Brent crude prices ($60–$80/bbl). Financial metrics: 9,371B KZT revenue, 2,393B KZT EBITDA, 0.16x Net Debt/EBITDA in 2025.	Capital expenditure must be strictly managed to preserve cash flow and support regional employment.	Predictive financial modeling & cash-flow stress testing using Big Data algorithms to optimize CAPEX allocation across operating assets.
Social & Regional Employment	KMG is the largest employer in Mangystau/Atyrau regions (>52k workforce). High social sensitivity around job security and local infrastructure.	Cannot aggressively cut headcount or close mature fields without triggering regional socio-economic instability.	Computer vision safety monitoring (TUMAR) to protect frontline workers and VR/AI workforce upskilling platforms.
Technological & Digital	Deployment of proprietary ABAI Big Data/AI platform, TUMAR safety computer vision, and Slim Hole drilling technology across subsidiaries.	Need to modernize legacy field infrastructure and transition from reactive maintenance to data-driven operations.	Scaling ABAI's 'Waterflooding Management' module (1,660 injection cells) and predictive failure analytics across all refineries.
Environmental & Decarbonization	Commitment to Low-Carbon Program 2060, OGMP 2.0 Gold Standard methane tracking, APG flaring utilization (98.8%), 247 MW Zhanaozen hybrid plant.	Pressure from international ESG investors to lower carbon intensity while maintaining national energy security.	Satellite & drone optical gas imaging (OGI) with AI leak detection algorithms to build an automated real-time 'Methane Leak Atlas'.
Global & Trade Supply Chain	Export dependency on Caspian Pipeline Consortium (CPC pipeline, 83M tonnes capacity) and cross-border transit through	Geopolitical Black Sea bottleneck risks and sanctions compliance mandate strict legal/operational	AI-based supply chain route optimization, predictive vessel scheduling (KMG

	Black Sea / Russia.	vigilance.	Systems & Services), and multi-route export balancing.

2. Artifact Week 2: Stakeholder and Ethics Map for KazMunayGas

This section categorizes KMG's internal and external stakeholder groups, outlining their operational claims, power/influence, and analyzing the core ethical and regulatory tensions inherent in operating as Kazakhstan's national energy champion.

Stakeholder Category	Key Entities & Profile	Core Expectations & Rights	Management & Engagement Approach
Internal: Owners & Shareholders	Samruk-Kazyna JSC (67.42%), Ministry of Finance RK (20.00%), National Bank RK (9.58%), Retail Free Float (3.00% on KASE/AIX).	Financial stability, dividend consistency (40–50% payout ratio), preservation of capital, strict corporate governance compliance.	Annual General Meetings, quarterly audited financial statements (PwC), KASE/AIX Issuer Days, IR Department reporting.
Internal: Workforce & Management	52,000+ employees across Corporate Center, operating subsidiaries (OMG, EMG, MMG, ANPZ, PNHZ) and technical institutes (KMG Engineering).	Fair living wages, occupational health & zero-harm safety, non-discrimination, career development, social support packages.	Collective bargaining agreements (70+ social benefits), TUMAR AI safety system, 'Korghau' hazard cards, internal Ombudsman service.
External Primary: International Partners	International IOCs in mega-consortia: Chevron, ExxonMobil, Shell, Eni, Lukoil, TotalEnergies, CNOOC, Sinopec.	Contractual compliance under PSAs/Joint Operating Agreements, predictable regulatory environment, fair cost recovery.	Joint Operating Committees, Board representations, joint technology pilots (Zhanaozen 247 MW plant with Eni, CCUS with Chevron/Shell).
External Primary: B2B Offtakers & Suppliers	Wholesale fuel buyers, industrial enterprises, local oilfield service providers, utility operators, international traders.	Reliable product delivery, high fuel specifications (e.g., Altay-45 winter diesel), transparent procurement via ISEZ portal.	Long-term off-take contracts, digital e-procurement portal (ISEZ), PKO vendor qualification audits, transparent SLA tracking.
External	Ministry of Energy RK,	Guaranteed domestic	Direct state reporting,

Secondary: Government & Regulators	Ministry of Ecology, State Revenue Committee (KGD), Agency for Protection of Competition (AZRK).	energy security, subsidized fuel supply, tax revenues, environmental compliance (Ecology Code, OGMP 2.0).	automated emission monitoring systems connected to Napor/MEPR, compliance reviews, regulatory dialogues.
External Secondary: Communities & Public	Local communities in Mangystau/Atyrau regions, general population of Kazakhstan, environmental NGOs.	Affordable domestic fuel prices, environmental protection (Caspian marine fauna), regional infrastructure investment.	Public ESG reporting (GRI/CDP), social infrastructure projects (Kenderli desalination plant, Zhanaozen Youth Palace), CSR initiatives.

Focal Ethical and Regulatory Tension Analysis for KMG

The central institutional tension facing KMG is its Dual Mandate Dilemma: balancing commercial efficiency and shareholder profitability against state-mandated social and regulatory obligations. As a public company listed on KASE/AIX, KMG is expected to maximize EBITDA, lower debt, and deliver competitive dividends to minority and sovereign shareholders. However, as the national energy operator, KMG is legally bound to sell crude oil to domestic refineries at subsidized prices well below world market export parity. Furthermore, KMG must absorb significant non-commercial social expenditures such as constructing the 121.2B KZT Kenderli desalination plant and maintaining employment at mature, high-cost fields in Mangystau to preserve regional socio-economic stability. KMG navigates this tension through structured portfolio cross-subsidization, leveraging high-margin export barrels from mega-projects (Tengiz, Kashagan, Karachaganak) to fund domestic supply obligations and regional infrastructure.

3. Official Course AI-Use Declaration (Level 2: Supported Use)

A. Overall AI-Use Status

[✓] I used one or more AI systems while preparing this assignment. I have documented each use below.

[ ] I did not use any AI system for this assignment.

B. AI Systems Register

AI System / Model	Provider / Version	Date(s) Used	Purpose in this Assignment
Gemini	Google AI (Gemini 2.5 Architecture)	September 18–20, 2026	IEEE referencing formatting and translation.

C. Detailed AI Activity and Verification Log

Part of Assignment / Activity	**AI System**	**What I Used AI To Do**	**How I Verified Output / What I Changed & Corrected**
Section 1: Business Context	Gemini	Helped organize and summarize 2025 financial and operational information for KMG and Shell from the sources I provided.	I independently checked all key figures against the companies' 2025 Annual Reports and financial statements. I corrected exchange-rate conversions and removed any unsupported figures.
Section 2: Environment & Stakeholders	Gemini	Helped me structure the PESTEL analysis, stakeholder groups, ownership information and key tensions into a clearer framework.	I verified ownership percentages and other quantitative claims using KMG's official Governance Report and Annual Report. I removed or corrected information that could not be supported by official sources.
Section 3: Business Model & AI Transformation	Gemini	Helped me organize KMG's digital initiatives (ABAI, TUMAR, Slim Hole, T-Pulse) and connect them with business-model components.	I independently checked the initiatives and their reported results against KMG's official Annual Report and supporting company information. I only kept claims that I could verify.
Sections 4 & 5: Transfer & Recommendations	Gemini	Used AI for brainstorming possible technology-transfer ideas and organizing	I evaluated the suggestions against KMG's actual business context, official evidence and

		recommendations and potential KPIs.	Kazakhstan's energy-security considerations. I made the final decisions and recommendations myself and rejected suggestions that were not appropriate for KMG.

D. Evidence and Quantitative-Claim Verification Log

Quantitative Claim / Figure	**Evidence Source**	**Independent Verification Performed**	**Result / Verification Status**
KMG 2025 Oil & Condensate Production: 26.211M tones (544 kbopd)	KMG 2025 Annual Report, p. 171 & Financial Statements	Checked the figure against KMG's official reporting and financial documentation.	VERIFIED
KMG 2025 Revenue: 9,371B KZT; EBITDA: 2,393B KZT	KMG 2025 Financial Statements, Note 6, p. 326	Checked the consolidated income statement figures against the official financial statements.	VERIFIED
KMG Net Debt: 375B KZT; Net Debt / EBITDA: 0.16x	KMG 2025 Financial Statements & Governance Report	Checked the reported debt figures and independently calculated the ratio using reported EBITDA.	VERIFIED / CALCULATED
Shell 2025 Hydrocarbon Production: 2.800M boe/d; Adjusted Earnings: $18.5B	Shell 2025 Annual Report, p. 385	Checked the figures against Shell's official 2025 reporting and consolidated financial statements.	VERIFIED
ABAI Waterflooding AI Coverage: 1,660 injection cells across OMG, MMG and KOA	KMG 2025 Annual Report, Digitalization section, p. 178	Checked the reported coverage against KMG's official documentation.	VERIFIED
Slim Hole Technology: 30%	KMG 2025 Annual Report, Production &	Checked the reported trial result against	VERIFIED

CAPEX reduction / 4.6B KZT savings at Uzen	Technology, p. 181	KMG's official reporting.	

E. Authorship and Judgment Reflection

One AI suggestion I did not accept, and why:
AI suggested considering Shell's asset-divestment and refinery-shutdown approach as a possible model for KMG. I did not use this as my recommendation because KMG operates within a different national energy-security and domestic-market context. I therefore evaluated the suggestion against KMG's specific responsibilities and Kazakhstan's energy sector rather than copying the AI-generated idea.

One material change I made after checking evidence independently:
I replaced broad, generic net-zero claims with Kazakhstan-specific and KMG-specific evidence that I could verify in official sources. My final discussion focuses on initiatives such as methane-emissions reduction, associated-gas flaring reduction and hybrid renewable/gas generation.

Remaining uncertainty or limitation:
The report identifies KMG's dependence on the Caspian Pipeline Consortium (CPC) and Black Sea transit routes as an external geopolitical and logistics risk. This is a limitation because the risk is influenced by factors outside KMG's direct operational control.

F. Assignment-Specific Confirmations & Final Declaration

I declare that this form accurately records my AI use for this assignment. I independently verified the claims, numbers and sources for which I remain accountable, and I authored the final analysis and judgment submitted for assessment. Any AI support listed above was used only as disclosed in this form.

[✓] Every quantitative claim in my report is traceable to evidence and independent verification.

[✓] My Company A / Company B transfer judgments (adopt, adapt, pilot, delay or reject) are my own reasoned analytical judgments.

[✓] My final recommendations are specific to KMG and the Kazakhstan context rather than copied from AI output.

[✓] My reference list contains only sources I independently checked and verified.

[✓] I confirm that I did not fabricate, invent, or knowingly submit unverifiable data, claims, or references.

[✓] I confirm that I did not upload prohibited course texts to any AI system.

Student Signature: Kenzhebay Ayaulym Date: 20.09.2026